$x_1+x_2+x_3+\cdots+x_n=r$, $x_i \geq 0$ C^{n+r-1}_{r}

非負整数 nonnegative interger solution

A B C D

ex: identical objects

↓

distinct container

Fundamental of Logic

- truth table
- equivalence
- inference

To prove $\sqrt{2}$ is irrational

<pf> Assume $\sqrt{2}$ 是有理数

$\sqrt{2}=\frac{p}{q}$ p.q 互質

- Statement (敘述)
  proposition (命題) → evaluated (either T or F)

p	$\neg p$
T	F
F	T

- connective
  - and $\wedge$
  - or $\vee$
  - not $\neg$
  - XOR $\veebar$
  - implication $\rightarrow$
  - bi-condition $(p\rightarrow q)\wedge(q\rightarrow p)$

    $p\leftrightarrow q$

    if and only if / p iff q

p	q	$p\wedge q$	$p\vee q$	$p\veebar q$	$p\rightarrow q$	$p\leftrightarrow q$
T	T	T	T	T	T	T
T	F	F	T	F	F	F
F	T	F	T	F	T	F
F	F	F	F	T	T	T

T F

p	$p\wedge\neg p$	$p\vee\neg p$
T	F	T
F	F	T

永遠T

↑ tautology

$(p\rightarrow q)\wedge(q\rightarrow p)$	$q\rightarrow p$
	T
	T
	F
	T

T
T
F
T

Contradiction

T T T
F T T
T F F
F F T

3/4 等價 / 真值表

1. (a) $[p \wedge (p \to q)] \to q$

(b) $\{(p \to q) \wedge (q \to r)] \to (p \to r)$

p	q	